# Evaluating the Performance of YOLO Object Detectors for Plant Disease Detection

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***Abstract*—Early and accurate detection of plant diseases is crucial for ensuring food security and minimizing crop losses. Deep learning-based object detection models, particularly those based on the You Only Look Once (YOLO) architecture, have shown promise in this area. This study compares the performance of five recent YOLO models - YOLOv5, YOLOv5-p6, YOLOv6, YOLOv8, and YOLOv9 - for plant disease detection using the PlantDoc dataset. The models were evaluated based on their accuracy, speed, and resource requirements. Our results indicate that YOLOv9-C and YOLOv8s demonstrated superior performance in terms of F1-Score, Recall, and mAP metrics. These models achieved an F1-Score of 0.39 and 0.37 respectively, with YOLOv9-C leading in Recall at 0.51 and both models tying at a Best mAP@50 of 46%. Moreover, the highest F1-Score was achieved by YOLOv5s at 0.42, indicating that this model also has potential for further exploration and optimization. The highest mAP50-95 was achieved by YOLOv8s at 36%, suggesting that this model may be particularly effective at detecting a wide range of plant diseases across different thresholds.**

***Index Terms*—Object detection, Precision agriculture, YOLO, Crop health monitoring, Computer vision, Deep learning**

## I. Introduction

Plant diseases pose a significant threat to global food security [1], causing substantial crop losses and economic damage. Early and accurate detection of diseases is crucial for implementing timely and effective management strategies to mitigate these losses and ensure food sustainability [2]. Traditional methods for disease detection rely on visual inspection by experts, which is time-consuming, subjective, and often impractical for large-scale applications. The development of deep learning models has revolutionized plant disease detection by offering automated, accurate, and efficient solutions [3]. These models can analyze large datasets of images with high accuracy and speed, surpassing the limitations of manual inspection [4]. In particular, object detection algorithms have proven highly effective in identifying and localizing disease symptoms on plants. Among object detection algorithms, one-stage detectors like the You Only Look Once (YOLO) [5] series offer significant advantages over two-stage detectors. One-stage detectors perform object detection and classification in a single step, resulting in faster inference speeds crucial for real-time applications. This is in contrast to two-stage detectors, which first generate region proposals and then classify them, leading to slower processing times. For example, YOLOv5 can achieve real-time detection speeds of over 100 frames per second (FPS), while a two-stage detector like Faster R-CNN [6] typically operates at around 10 FPS. This speed advantage makes YOLO models particularly suitable for deployment on resource-constrained devices in the field. This study presents a comparative analysis of five recent YOLO models for plant disease detection: YOLOv5, YOLOv5-P6, YOLOv6, YOLOv8, and YOLOv9. These models were evaluated on the PlantDoc dataset, which contains images of various plant species with diverse disease and healthy conditions. This study aims to provide valuable insights for researchers and practitioners in selecting the optimal YOLO model for plant disease detection based on their specific requirements and constraints. By analyzing the performance of these models across different metrics, this study can guide the development and deployment of efficient and accurate plant disease detection systems, contributing to improved crop health and global food security.

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## II. YOLO (You Only Look Once) and Frameworks

YOLO (You Only Look Once) represents a seminal framework in the realm of real-time object detection. Unlike its

predecessors, YOLO reframes object detection as a regression problem, directly predicting bounding boxes and class probabilities from full images within a single evaluation [7]. This unified approach circumvents the complexities of traditional multi-stage object detectors. While YOLO can be implemented within various frameworks, Darknet was the most common framework for versions prior to YOLOv5 [8]. Successive iterations of YOLO (from YOLOv5 to YOLOv9) have demonstrated continuous evolution and refinement. These enhancements often focus on architectural modifications, novel loss functions, and advanced training techniques. YOLO's speed and adaptability have propelled its adoption across diverse applications, including autonomous vehicles, medical image analysis, and precision agriculture [9] [10].

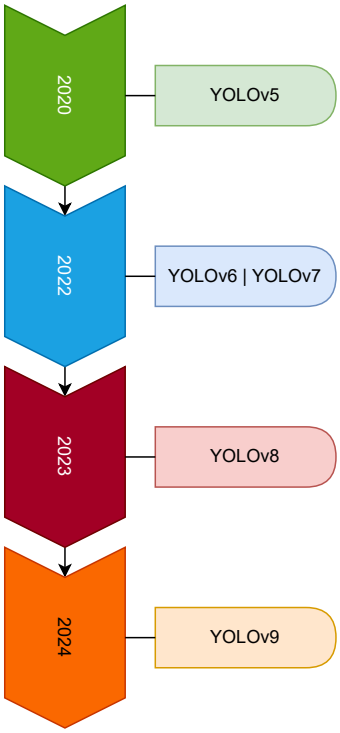



Fig. 1. YOLO Evolution

### A. YOLOv5

YOLOv5 [11] built upon the strengths of its predecessors while offering greater flexibility and ease of use. It provided various model sizes (Nano, Small, Medium, Large, XLarge) to cater to different speed-accuracy requirements and adopted the PyTorch framework for easier implementation and modification. YOLOv5 also implemented EfficientDet [12] inspired scaling rules for adjusting model size and employed the YOLOv3 head for objectness and classification prediction. Due to its ease of use, flexibility, and performance, YOLOv5 became widely adopted within the computer vision community.

### B. YOLOv6

YOLOv6 [13] focused on deployment efficiency while maintaining competitive performance. It introduced the RepVGG backbone [14], which utilizes re-parameterization for efficient inference, and replaced the YOLOv3 head with a decoupled head for objectness and classification. Additionally, YOLOv6 utilized SimOTA label assignment for improved target selection and employed various training techniques like head decoupling and auxiliary head to boost accuracy.

### C. YOLOv8

YOLOv8 [15] introduced a unified architecture for object detection and instance segmentation, employing task-specific heads for different tasks while sharing a common backbone and neck. It utilized an anchor-free detection head for efficient inference and offered various model sizes for different speed-accuracy trade-offs. YOLOv8 achieved competitive performance in both object detection and instance segmentation tasks.

### D. YOLOv9

YOLOv9 [16] focused on addressing information loss during the feedforward process in deep networks. It introduced the concept of programmable gradient information (PGI) to generate reliable gradients and designed the Generalized Efficient Layer Aggregation Network (GELAN) for improved parameter utilization and accuracy. Notably, YOLOv9 utilized conventional convolution operators to achieve better performance than depth-wise convolution based designs. It achieved state-of-the-art performance on the MS COCO dataset, surpassing previous train-from-scratch methods.

## III. Materials and Methods

This section will explore the materials and methods employed in our study. Specifically, we will discuss the dataset used, the training process, and the evaluation metrics. These details are crucial for understanding the foundation of our research and the methodology behind our YOLO-based plant disease recognition model.

### A. Data Collection and Labelling

The PlantDoc dataset [17] is a valuable resource for developing visual plant disease detection systems. To optimize its performance, we leveraged a modified version enhanced by the Roboflow team [18]. These enhancements included the correction of 28 annotation inconsistencies, the manual labeling of five previously unlabeled images, and the provision of format conversion capabilities for seamless integration with various YOLO versions and other machine learning frameworks. These modifications augmented the original PlantDoc dataset—comprising 2669 images across 13 plant species and up to 17 disease classes representing approximately 300 human hours of annotation effort—thereby

TABLE I
QUANTITATIVE RESULTS OF THE TRAINING PROCESS.

| Models | F1-Score | Recall | Last mAP@50 | Best mAP@50 | mAP50-95 |
|---|---|---|---|---|---|
| Yolov5s | 0.42 | 0.49 | 45% | 46% | 34% |
| Yolov5s6 | 0.28 | 0.42 | 34% | 34% | 26% |
| Yolov6-s | 0.18 | 0.39 | 23% | 23% | 17% |
| Yolov8s | 0.37 | 0.45 | 43.4% | 43.6% | 33% |
| Yolov9-C | 0.39 | 0.51 | 45% | 46% | 36% |

creating a more robust foundation for training plant disease detection models.

TABLE II
TRAIN AND VALIDATION COMPOSITION OF THE DATASET

| Set | Size |
|---|---|
| Train | 2330 |
| Validation | 239 |
| Totale | 2669 |

### *B. Training, Validation and performance metrics*

The YOLO models (v5, v5-p6, v6, v7, v8 and v9) were subjected to a training process utilizing the dataset described in this study. This dataset was partitioned into a training set and a validation set as described in table II to ensure robust model training and evaluation. The training process for each model was executed employing the pre-trained weights given by the developers of YOLO. The YOLO configuration file facilitates the alteration of hyperparameters. However, in the context of this study, we opted to employ the default hyperparameters across all models.

In the process of evaluating the research, a variety of performance metrics were employed, each of which is commonly utilized in detection tasks such as the recognition of plant diseases.

- **Recall,** also referred to as the Detection Rate or Sensitivity. This metric gauges the model's capacity to correctly classify images into different predefined classes or categories 1.

$$\begin{aligned} \text{Recall} &= \frac{\text{True Positives}}{\text{Total Number of Actual Positives}} \\ &= \frac{TP}{TP + FN} \end{aligned} \tag{1}$$

- **F-measure,** F-Score, or F1 score was employed. This is the harmonic mean of recall and precision. It ranges from 0 as the worst value to 1 as the best. Using this metric is advantageous when classes are imbalanced and false negatives have serious consequences 2.

$$\mathbf{F1}_{\text{score}} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

- **mAP@50,** stands for Mean Average Precision at an Intersection over Union (IoU) threshold of 0.50, is frequently employed in object detection assignments. This metric quantifies the model's average precision when the IoU threshold is set to 0.50. In other words, any predictions made by the model that achieve an IoU of 0.50 or more with the ground truth are deemed accurate 3.

$$\text{mAP@50} = \frac{1}{|Q|} \sum_{q \in Q} \max_{d \in D} P(q, d) \cdot \text{IoU}(q, d) \geq 0.50 \tag{3}$$

- **mAP@50:95** This metric computes the Mean Average Precision over a spectrum of Intersection over Union (IoU) thresholds, ranging from 0.50 to 0.95, with an increment of 0.05 at each step. It offers a more holistic assessment of the model's performance by considering various degrees of overlap between the predicted and actual bounding boxes. This metric is especially valuable for gauging the model's effectiveness in situations that demand a high level of precision in object localization 4.

$$\begin{aligned} \text{mAP@50:95} = \frac{1}{|Q|} \sum_{q \in Q} \max_{d \in D} \\ \begin{cases} 1, & \text{if } 0.50 \leq \text{IoU}(q, d) \leq 0.95 \\ 0, & \text{otherwise} \end{cases} \end{aligned} \tag{4}$$

These metrics collectively provide a comprehensive evaluation of the model's performance across various aspects, ensuring a robust assessment of its effectiveness in recognizing plant diseases. They facilitate the identification of areas where the model excels, as well as those where improvements may be needed. This, in turn, can guide future efforts to refine and optimize the model. This rigorous evaluation methodology underscores the academic rigor of this research.

## IV. RESULTS

### *A. Evaluating the Precision of Predictions on the Validation Set*

The F1-score comparison in Figure 2 shows that Yolov9-C leads with a score of 0.39, indicating its superior balance in identifying true positives while minimizing false positives and negatives. This is followed by Yolov8s and Yolov5s. The

TABLE III
TRAINING DETAILS AND HYPERPARAMETERS.

| Models | Epoch | Framework | Billions of Flops |
|---|---|---|---|
| Yolov5s | 75 | PyTorch | 13.2 |
| Yolov5s6 | 75 | PyTorch | 24.6 |
| Yolov6-s | 75 | PyTorch | 45.3 |
| Yolov8s | 75 | PyTorch | 28.6 |
| Yolov9-C | 75 | PyTorch | 102.8 |

F1-score is a measure of a model's accuracy and considers both precision and recall. The lower score of Yolov6s at 0.18 suggests that there is room for improvement in its precision-recall balance.

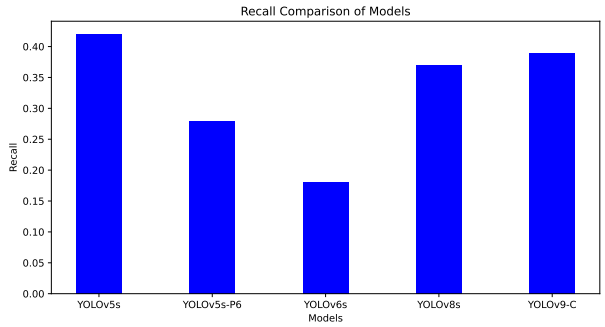



Fig. 2. F1-score comparison using the validation dataset

In the mAP@50 and mAP@50:95 comparison, shown in Figure 3 both Yolov8s and Yolov9-C are tied at the top with a 46% score each. This metric assesses the Mean Average Precision at an Intersection over Union (IoU) threshold of 0.50, which mirrors each model's capability to accurately detect objects with a substantial overlap between the predicted and actual bounding boxes. The diminished scores of other models, such as Yolov6s at 23%, suggest a lower accuracy in object detection at this specific IoU threshold. This is indicative of the model's performance in scenarios that require high precision in object detection.

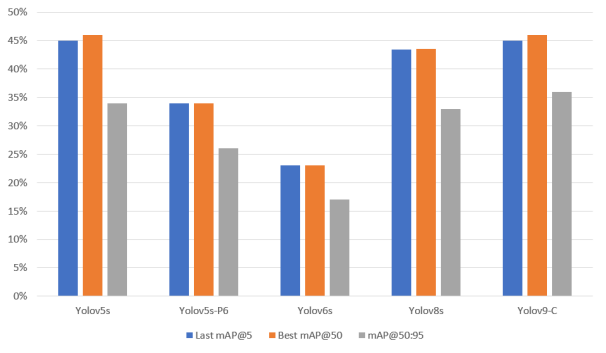



Fig. 3. mAP@50 and mAP@50:95 comparison

The recall comparison graph in Figure 4 shows that Yolov9-C stands out with a leading score of 0.51, indicating that it identifies over half of all relevant objects effectively. Recall measures the model's ability to identify all relevant instances. The second highest is Yolov5s at 0.49 recall value indicating its effectiveness but still slightly less competent than Yolov9-C

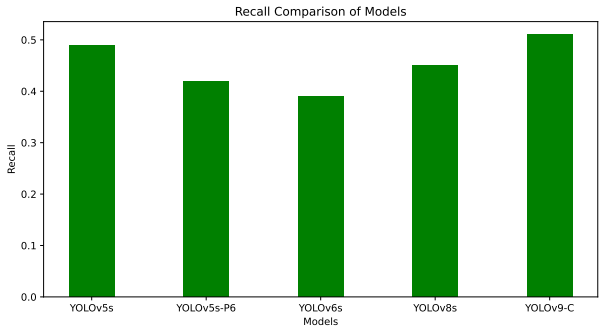



Fig. 4. Recall comparison

### *B. Precision-Recall Graphs and Optimal Confidence Threshold Explanation*

The Best Confidence Threshold in the context of object detection models refers to the optimal point on the Precision-Recall curve that yields the highest F1 score. The F1 score is the harmonic mean of precision and recall, and it measures the balance between these two metrics. The calculation of the area beneath the curve was executed utilizing the mAP@50 metric. As illustrated in Figure 4B, the confidence thresholds underwent optimization and adjustment to ensure the F-1 score for each model was at its peak. This process underscores the importance of fine-tuning model parameters to achieve optimal performance.Following this optimization process, the optimal confidence thresholds were determined to be 0.20, 0.11, 0.07, 0.31, and 0.18 for the v5, v5-p6, v6, v8 and v9 models, respectively 5.

Precision-recall curves [19] provide a visual representation of the trade-off between precision and recall at different decision thresholds. A model with high precision but low recall may be conservative in its predictions, while a model with high recall but low precision may be more aggressive in identifying diseased plants. Figure 6 shows the precision–recall curves for each trained YOLO model.

## V. DISCUSSION

The benchmark results for plant disease recognition using various YOLO models on the PlantDoc dataset provide insightful observations. Among the evaluated models, the YOLOv9-C and YOLOv8s models have demonstrated superior performance in terms of F1-Score, Recall, and mAP metrics. Specifically, YOLOv9-C achieved an F1-Score of 0.39 and a leading Recall of 0.51, while both YOLOv9-C and YOLOv8s tied for the highest Best mAP@50 of 46%.

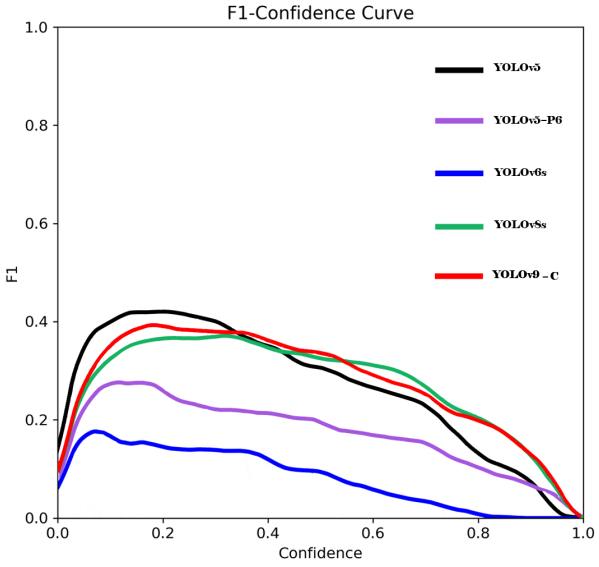



Fig. 5. Best Confidence Threshold

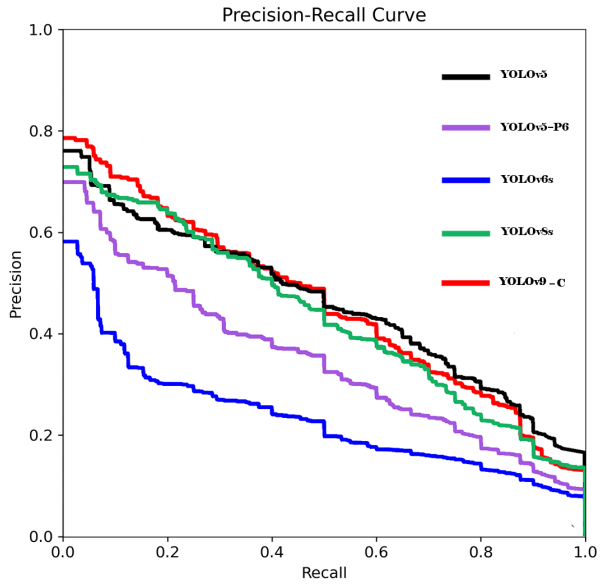



Fig. 6. Precision-Recall curves

These metrics indicate a promising direction for enhancing the accuracy and efficiency of plant disease identification using these advanced YOLO models.

Interestingly, the YOLOv5s model attained the highest overall F1-Score at 0.42, highlighting its potential for further exploration and optimization. Additionally, the highest mAP50-95 was recorded by YOLOv8s at 36%, suggesting that this model is particularly effective at detecting a wide range of plant diseases across different thresholds. This versatility across varying mAP thresholds underscores YOLOv8s's robustness in diverse detection scenarios.

These results collectively suggest that while YOLOv9-C and YOLOv8s excel in certain metrics, YOLOv5s also shows considerable promise, particularly in achieving a higher F1-Score. This variability across different models indicates that the choice of YOLO architecture may depend on specific application needs and performance priorities.

## VI. Future Directions

Future research could focus on several key areas to build upon these findings:

- **Model Optimization**:
  - *Architecture Exploration*: Further investigation into alternative architectures and modifications to the YOLO framework could yield improvements in detection performance. This could include experimenting with different layers, activation functions, and hyperparameter tuning.
  - *Feature Integration*: Incorporating additional features such as multi-spectral imaging data or environmental factors (e.g., humidity, temperature) could enhance model accuracy. The integration of domain-specific knowledge about plant diseases might also contribute to better performance.
- **Dataset Expansion**:
  - *Diverse Plant Species*: Expanding the dataset to include a broader range of plant species and disease types can improve the model's generalization capabilities. Ensuring a balanced representation of different disease severities and stages is crucial.
  - *High-Resolution Imagery*: Utilizing higher resolution images might help in capturing finer details, aiding in the detection of early-stage diseases which are often more challenging to identify.
- **Real-Time Application**:
  - *Field Deployment*: Developing lightweight, real-time versions of these models suitable for deployment on mobile devices or drones can facilitate early detection in agricultural settings. This would enable farmers to take timely action, thereby minimizing crop loss and enhancing food security.
  - *Automated Monitoring Systems*: Integrating these models into automated monitoring systems can provide continuous surveillance of crops, leading to more proactive and precise disease management.
- **Transfer Learning and Domain Adaptation**:
  - *Transfer Learning*: Applying transfer learning techniques to leverage pre-trained models on larger, more diverse datasets can enhance the detection performance on specific plant disease datasets like PlantDoc.

  - *Domain Adaptation*: Adapting models trained on synthetic or related datasets to the target domain (real-world agricultural settings) can improve their robustness and applicability.
- **Collaborative Approaches**:
  - *Interdisciplinary Collaboration*: Collaborating with plant pathologists, agronomists, and data scientists can provide a multidisciplinary approach to refining these models. This collaboration can ensure that the models are both scientifically accurate and practically useful.

## VII. Conclusion

These benchmark results provide a valuable foundation for future research in plant disease recognition using YOLO models. By continuing to refine and optimize these models, we can strive towards more accurate, efficient, and real-time plant disease detection, contributing to sustainable agriculture and food security. The findings provide valuable guidance for selecting the optimal YOLO model for plant disease detection based on specific needs and constraints. Future research can explore further optimization and customization of these models for specific plant species and diseases. This endeavor aligns with the broader objective of harnessing the power of artificial intelligence to address pressing global challenges.

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